

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.utils import to_categorical
import matplotlib.pyplot as plt
import numpy as np
```

```
import tensorflow_datasets as tfds

print("Loading EMNIST Letters to dataset...")
(ds_train, ds_test), ds_info = tfds.load(
    'emnist/letters',
    split=['train', 'test'],
    shuffle_files=True,
    as_supervised=True,
    with_info=True,
)

def preprocess_emnist(image, label):
    image = tf.cast(image, tf.float32) / 255.0
    image = tf.image.transpose(image)
    return image, label - 1

train_data = ds_train.map(preprocess_emnist).batch(32).prefetch(tf.data.AUTOTUNE)
test_data = ds_test.map(preprocess_emnist).batch(32).prefetch(tf.data.AUTOTUNE)
```

Loading EMNIST Letters to dataset...

WARNING:absl:Variant folder /root/tensorflow_datasets/emnist/letters/3.1.0 has no dataset_info.json

Downloading and preparing dataset Unknown size (download: Unknown size, generated: Unknown size, total: Unknown size) to /root/tensorflow_datasets/emnist/letters/3.1.0

DI Completed...: 100% 1/1 [00:11<00:00, 7.89s/ url]

DI Size...: 100% 535/535 [00:11<00:00, 70.75 MiB/s]

Extraction completed...: 100% 30/30 [00:11<00:00, 11.16s/ file]

Extraction completed...: 100% 4/4 [00:00<00:00, 4.00 file/s]

Dataset emnist downloaded and prepared to /root/tensorflow_datasets/emnist/letters/3.1.0. Subsequent calls will reuse this c

```
with tf.device('/GPU:0'):
    model = Sequential([
        tf.keras.layers.Flatten(input_shape=(28,28,1)),
        Dense(256, activation='relu'),
        tf.keras.layers.Dropout(0.2),
        Dense(128, activation='relu'),
        Dense(26, activation='softmax')
    ])
    model.compile(
        optimizer='adam',
        loss='sparse_categorical_crossentropy',
        metrics=['accuracy']
    )
    print("\nStarting training on 26 classes....")
    history = model.fit(
        train_data,
        epochs=10,
        validation_data=test_data
    )
```

Starting training on 26 classes....

Epoch 1/10

/usr/local/lib/python3.12/dist-packages/keras/src/layers/reshaping/flatten.py:37: UserWarning: Do not pass an `input\_shape` /

super().__init__(**kwargs)

2775/2775 — 32s 11ms/step - accuracy: 0.6662 - loss: 1.1262 - val_accuracy: 0.8406 - val_loss: 0.4821

Epoch 2/10

2775/2775 — 26s 9ms/step - accuracy: 0.8479 - loss: 0.4797 - val_accuracy: 0.8703 - val_loss: 0.3981

Epoch 3/10

2775/2775 — 41s 9ms/step - accuracy: 0.8740 - loss: 0.3903 - val_accuracy: 0.8764 - val_loss: 0.3816

Epoch 4/10

2775/2775 — 25s 9ms/step - accuracy: 0.8859 - loss: 0.3476 - val_accuracy: 0.8849 - val_loss: 0.3557

Epoch 5/10

2775/2775 — 25s 9ms/step - accuracy: 0.8943 - loss: 0.3184 - val_accuracy: 0.8878 - val_loss: 0.3530

Epoch 6/10

2775/2775 — 24s 9ms/step - accuracy: 0.9016 - loss: 0.2920 - val_accuracy: 0.8878 - val_loss: 0.3474

Epoch 7/10

2775/2775 — 40s 8ms/step - accuracy: 0.9038 - loss: 0.2788 - val_accuracy: 0.8861 - val_loss: 0.3582

Epoch 8/10

2775/2775 — 23s 8ms/step - accuracy: 0.9079 - loss: 0.2655 - val_accuracy: 0.8922 - val_loss: 0.3519

Epoch 9/10

2775/2775 — 24s 9ms/step - accuracy: 0.9140 - loss: 0.2512 - val_accuracy: 0.8963 - val_loss: 0.3344

Epoch 10/10

2775/2775 — 25s 9ms/step - accuracy: 0.9144 - loss: 0.2437 - val_accuracy: 0.8935 - val_loss: 0.3499

```
def get_letter(label):
    return chr(ord('A')+label)
images, labels= next(iter(test_data))
predictions = model.predict(images)
plt.figure(figsize=(15,4))
for i in range(10):
    plt.subplot(1,10,i+1)
    plt.imshow(images[i].numpy().squeeze(), cmap='gray')
    pred_idx = np.argmax(predictions[i])
    true_idx = labels[i].numpy()
    color='green' if pred_idx == true_idx else 'red'
    plt.title(f"p: {get_letter(pred_idx)} \nT :{get_letter(true_idx})", color=color)
    plt.axis('off')

plt.tight_layout
plt.show()
```

1/1 ————— 0s 137ms/step



```
loss, accuracy = model.evaluate(test_data, verbose=0)
print(f"\nFinal Test Accuracy for Letters: {accuracy*100:.2f}%")
```

Final Test Accuracy for Letters: 89.35%